Joining, assembly & finishing

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe mortise and tenon, lap, rub, halving, dowel, rebate and housing joints and where they are used
- ☐ choose between PVA and epoxy resin, screws, nails, nuts and bolts, pop rivets and welding
- ☐ describe how to prepare and apply each surface finish: sanding, polishing, varnish, oil, stain, wax, paint, dip coating
- ☐ describe how to finish the edges of acrylic

I DO — Worked example

Read how a full-mark answer is built

A clear acrylic phone stand has been cut out with a coping saw.

1 Describe how the edges of the acrylic would be finished to a high quality. (3 marks)

Full-mark answer each ✓ is one mark

Draw file the edge to remove the saw marks, keeping the file flat ✓. Rub down with wet and dry paper, moving through finer grades ✓. Polish the edge with acrylic polish and a soft cloth to a clear finish ✓.

Why marks are not awarded

"File it, then sand it with sandpaper, then use steel wool."

1 mark: filing scores, but sandpaper and steel wool are both abrasives and count as one at best; glass paper is not suitable for acrylic.

WE DO — Guided practice

Use the starters and word bank to help you

Holes are drilled in the acrylic before it is bent.

2 Explain why the holes are drilled before bending. (1 mark)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

accurately · easier · position · Acrylic · support · cracking · chance

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which joint is best for fixing shelves into the sides of a bookcase?
☐ Housing ☐ Halving ☐ Mortise and tenon ☐ Lap

b) Why are knock-down fittings used in flat-pack furniture?
☐ They are permanent, so the product can't be taken apart ☐ Product can be packed flat and assembled at home ☐ They are stronger than any traditional wood joint ☐ They hide every screw, so the product looks seamless

c) What is the purpose of a pilot hole for a screw?
☐ To guide the screw and stop the wood splitting ☐ To let the screw head sit below the surface ☐ To make the screw easy to remove later ☐ To hold extra glue around the screw

d) Before varnishing wood you should…
☐ Sand across the grain with a coarse paper first ☐ Wet the surface and varnish while it is still damp ☐ Apply a coat of oil to seal the grain beforehand ☐ Sand with the grain through finer grades and remove dust

4 Key terms: write the letter of the matching description next to each term.

1. **Mortise and tenon**	____	**A** Two-part adhesive that joins different materials.
2. **Housing joint**	____	**B** Carcase joint for shelves in bookcases and cabinets.
3. **Knock-down fitting**	____	**C** Non-permanent fitting for flat-pack furniture.
4. **Epoxy resin**	____	**D** Strong frame joint for tables, chairs and doors.
5. **Pop rivet**	____	**E** Hot metal dipped in plastic powder: protective, coloured, grippy finish.
6. **Dip coating**	____	**F** Joins thin sheet from one side only.

A copper key fob is joined to a beech handle.

5 Name a suitable adhesive and explain why it is suitable. (2 marks)

The acrylic phone stand needs a single 60° bend.

6 Describe how the acrylic would be bent accurately. (3 marks)

An oak coffee table is to be finished with varnish.

7 Describe how a high quality brushed varnish finish would be achieved. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?